1

Experiments with random outcomes

The purpose of probability theory is to build mathematical models of experiments with random outcomes and then analyze these models. A random outcome is anything we cannot predict with certainty, such as the flip of a coin, the roll of a die, the gender of a baby, or the future value of an investment.

1.1. Sample spaces and probabilities

The mathematical model of a random phenomenon has standard ingredients. We describe these ingredients abstractly and then illustrate them with examples.

Definition 1.1. These are the ingredients of a probability model.

- The **sample space** Ω is the set of all the possible outcomes of the experiment. Elements of Ω are called **sample points** and typically denoted by ω.
- Subsets of Ω are called **events**. The collection of events in Ω is denoted by $\mathcal{F}$. ♣
- The **probability measure** (also called **probability distribution** or simply **probability**) P is a function from $\mathcal{F}$ into the real numbers. Each event A has a probability $P(A)$, and P satisfies the following axioms.
  (i) $0 \le P(A) \le 1$ for each event A.
  (ii) $P(\Omega) = 1$ and $P(\varnothing) = 0$.
  (iii) If $A_1, A_2, A_3, \ldots$ is a sequence of pairwise disjoint events then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i). \tag{1.1}$$

The triple $(\Omega, \mathcal{F}, P)$ is called a **probability space**. Every mathematically precise model of a random experiment or collection of experiments must be of this kind.

The three axioms related to the probability measure P in Definition 1.1 are known as *Kolmogorov's axioms* after the Russian mathematician Andrey Kolmogorov who first formulated them in the early 1930s.

A few words about the symbols and conventions. Ω is an upper case omega, and ω is a lower case omega. $\varnothing$ is the empty set, that is, the subset of Ω that contains no sample points. The only sensible value for its probability is zero. *Pairwise disjoint* means that $A_i \cap A_j = \varnothing$ for each pair of indices $i \neq j$. Another way to say this is that the events A_i are *mutually exclusive*. Axiom (iii) says that the probability of the union of mutually exclusive events is equal to the sum of their probabilities. Note that rule (iii) applies also to finitely many events.

Fact 1.2. If $A_1, A_2, \dots, A_n$ are pairwise disjoint events then

$$P(A_1 \cup \cdots \cup A_n) = P(A_1) + \cdots + P(A_n). \tag{1.2}$$

Fact 1.2 is a consequence of (1.1) obtained by setting $A_{n+1} = A_{n+2} = A_{n+3} = \cdots = \varnothing$. If you need a refresher on set theory, see Appendix B.

Now for some examples.

Example 1.3. We flip a fair coin. The sample space is $\Omega = \{\mathtt{H}, \mathtt{T}\}$ ($\mathtt{H}$ for heads and $\mathtt{T}$ for tails). We take $\mathcal{F} = \{\varnothing, \{\mathtt{H}\}, \{\mathtt{T}\}, \{\mathtt{H}, \mathtt{T}\}\}$, the collection of all subsets of Ω. The term "fair coin" means that the two outcomes are equally likely. So the probabilities of the singletons $\{\mathtt{H}\}$ and $\{\mathtt{T}\}$ are

$$P\{\mathtt{H}\} = P\{\mathtt{T}\} = \tfrac{1}{2}.$$

By axiom (ii) in Definition 1.1 we have $P(\varnothing) = 0$ and $P\{\mathtt{H}, \mathtt{T}\} = 1$. Note that the "fairness" of the coin is an assumption we make about the experiment. ▲

Example 1.4. We roll a standard six-sided die. Then the sample space is $\Omega = \{1, 2, 3, 4, 5, 6\}$. Each sample point ω is an integer between 1 and 6. If the die is fair then each outcome is equally likely, in other words

$$P\{1\} = P\{2\} = P\{3\} = P\{4\} = P\{5\} = P\{6\} = \tfrac{1}{6}.$$

A possible event in this sample space is

$$A = \{\text{the outcome is even}\} = \{2, 4, 6\}. \tag{1.3}$$

Then

$$P(A) = P\{2, 4, 6\} = P\{2\} + P\{4\} + P\{6\} = \tfrac{1}{2}$$

where we applied Fact 1.2 in the second equality. ▲

Some comments about the notation. In mathematics, sets are typically denoted by upper case letters A, B, etc., and so we use upper case letters to denote events. Like A in (1.3), events can often be expressed both in words and in mathematical symbols. The description of a set (or event) in terms of words or mathematical symbols is enclosed in braces $\{\ \}$. Notational consistency would seem to require that the probability of the event $\{2\}$ be written as $P(\{2\})$. But it seems unnecessary to add the parentheses around the braces, so we simplify the expression to $P\{2\}$ or $P(2)$.

Example 1.5. (Continuation of Examples 1.3 and 1.4) The probability measure P contains our assumptions and beliefs about the phenomenon that we are modeling.

If we wish to model a flip of a biased coin we alter the probabilities. For example, suppose we know that heads is three times as likely as tails. Then we define our probability measure P_1 by $P_1\{\mathtt{H}\} = \frac{3}{4}$ and $P_1\{\mathtt{T}\} = \frac{1}{4}$. The sample space is again $\Omega = \{\mathtt{H}, \mathtt{T}\}$ as in Example 1.3, but the probability measure has changed to conform with our assumptions about the experiment.

If we believe that we have a loaded die and a six is twice as likely as any other number, we use the probability measure $\widetilde{P}$ defined by

$$\widetilde{P}\{1\} = \widetilde{P}\{2\} = \widetilde{P}\{3\} = \widetilde{P}\{4\} = \widetilde{P}\{5\} = \tfrac{1}{7} \quad \text{and} \quad \widetilde{P}\{6\} = \tfrac{2}{7}.$$

Alternatively, if we scratch away the five from the original fair die and turn it into a second two, the appropriate probability measure is

$$Q\{1\} = \tfrac{1}{6},\ Q\{2\} = \tfrac{2}{6},\ Q\{3\} = \tfrac{1}{6},\ Q\{4\} = \tfrac{1}{6},\ Q\{5\} = 0,\ Q\{6\} = \tfrac{1}{6}. \qquad \blacktriangle$$

These examples show that to model different phenomena it is perfectly sensible to consider different probability measures on the same sample space. Clarity might demand that we distinguish different probability measures notationally from each other. This can be done by adding ornaments to the P, as in P_1 or $\widetilde{P}$ (pronounced "P tilde") above, or by using another letter such as Q. Another important point is that it is perfectly valid to assign a probability of zero to a nonempty event, as with Q above.

Example 1.6. Let the experiment consist of a roll of a pair of dice (as in the games of Monopoly or craps). We assume that the dice can be distinguished from each other, for example that one of them is blue and the other one is red. The sample space is the set of pairs of integers from 1 through 6, where the first number of the pair denotes the number on the blue die and the second denotes the number on the red die:

$$\Omega = \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\}.$$

Here (a, b) is a so-called *ordered pair* which means that outcome $(3, 5)$ is distinct from outcome $(5, 3)$. (Note that the term "ordered pair" means that order matters, not that the pair is in increasing order.) The assumption of fair dice would dictate equal probabilities: $P\{(i, j)\} = \frac{1}{36}$ for each pair $(i, j) \in \Omega$. An example of an event of interest would be

$$D = \{\text{the sum of the two dice is 8}\} = \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$$

and then by the additivity of probabilities

$$P(D) = P\{(2,6)\} + P\{(3,5)\} + P\{(4,4)\} + P\{(5,3)\} + P\{(6,2)\}$$
$$= \sum_{(i,j):i+j=8} P\{(i,j)\} = 5 \cdot \tfrac{1}{36} = \tfrac{5}{36}. \quad \blacktriangle$$

Example 1.7. We flip a fair coin three times. Let us encode the outcomes of the flips as 0 for heads and 1 for tails. Then each outcome of the experiment is a sequence of length three where each entry is 0 or 1:

$$\Omega = \{(0,0,0), (0,0,1), (0,1,0), \ldots, (1,1,0), (1,1,1)\}. \tag{1.4}$$

This Ω is the set of ordered triples (or 3-tuples) of zeros and ones. Ω has $2^3 = 8$ elements. (We review simple counting techniques in Appendix C.) With a fair coin all outcomes are equally likely, so $P\{\omega\} = 2^{-3}$ for each $\omega \in \Omega$. An example of an event is

$$B = \{\text{the first and third flips are heads}\} = \{(0,0,0), (0,1,0)\}$$

with

$$P(B) = P\{(0,0,0)\} + P\{(0,1,0)\} = \tfrac{1}{8} + \tfrac{1}{8} = \tfrac{1}{4}. \quad \blacktriangle$$

Much of probability deals with repetitions of a simple experiment, such as the roll of a die or the flip of a coin in the previous two examples. In such cases *Cartesian product spaces* arise naturally as sample spaces. If $A_1, A_2, \ldots, A_n$ are sets then the Cartesian product

$$A_1 \times A_2 \times \cdots \times A_n$$

is defined as the set of ordered n-tuples with the ith element from A_i. In symbols

$$A_1 \times A_2 \times \cdots \times A_n = \{(x_1, \ldots, x_n) : x_i \in A_i \text{ for } i = 1, \ldots, n\}.$$

In terms of product notation, the sample space of Example 1.6 for a pair of dice can be written as

$$\Omega = \{1,2,3,4,5,6\} \times \{1,2,3,4,5,6\}$$

while the space for three coin flips in Example 1.7 can be expressed as

$$\Omega = \{0,1\} \times \{0,1\} \times \{0,1\} = \{0,1\}^3.$$

1.2. Random sampling

Sampling is choosing objects randomly from a given set. It can involve repeated choices or a choice of more than one object at a time. Dealing cards from a deck is an example of sampling. There are different ways of setting up such experiments which lead to different probability models. In this section we discuss three sampling mechanisms that lead to equally likely outcomes. This allows us to compute probabilities by counting. The required counting methods are developed systematically in Appendix C.

Before proceeding to sampling, let us record a basic fact about experiments with equally likely outcomes. Suppose the sample space Ω is a finite set and let $\#\Omega$ denote the total number of possible outcomes. If each outcome ω has the same probability then $P\{\omega\} = \frac{1}{\#\Omega}$ because probabilities must add up to 1. In this case probabilities of events can be found by counting. If A is an event that consists of elements $a_1, a_2, \dots, a_r$, then additivity and $P\{a_i\} = \frac{1}{\#\Omega}$ imply

$$P(A) = P\{a_1\} + P\{a_2\} + \cdots + P\{a_r\} = \frac{\#A}{\#\Omega}$$

where we wrote $\#A$ for the number of elements in the set A.

Fact 1.8. If the sample space Ω has finitely many elements and each outcome is equally likely then for any event $A \subset \Omega$ we have

$$P(A) = \frac{\#A}{\#\Omega}. \tag{1.5}$$

Look back at the examples of the previous section to check which ones were of the kind where $P\{\omega\} = \frac{1}{\#\Omega}$.

Remark 1.9. (Terminology) It should be clear by now that random outcomes do not have to be equally likely. (Look at Example 1.5 in the previous section.) However, it is common to use the phrase "an element is chosen at random" to mean that all choices are equally likely. The technically more accurate phrase would be "chosen *uniformly* at random." Formula (1.5) can be expressed by saying "when outcomes are equally likely, the probability of an event equals the number of favorable outcomes over the total number of outcomes." ▲

We turn to discuss sampling mechanisms. An *ordered sample* is built by choosing objects one at a time and by keeping track of the order in which these objects were chosen. After each choice we either replace (put back) or discard the just chosen object before choosing the next one. This distinction leads to *sampling with replacement* and *sampling without replacement*. An *unordered sample* is one where only the identity of the objects matters and not the order in which they came.

We discuss the sampling mechanisms in terms of an urn with numbered balls. An urn is a traditional device in probability (see Figure 1.1). You cannot see the contents of the urn. You reach in and retrieve one ball at a time without looking. We assume that the choice is uniformly random among the balls in the urn.

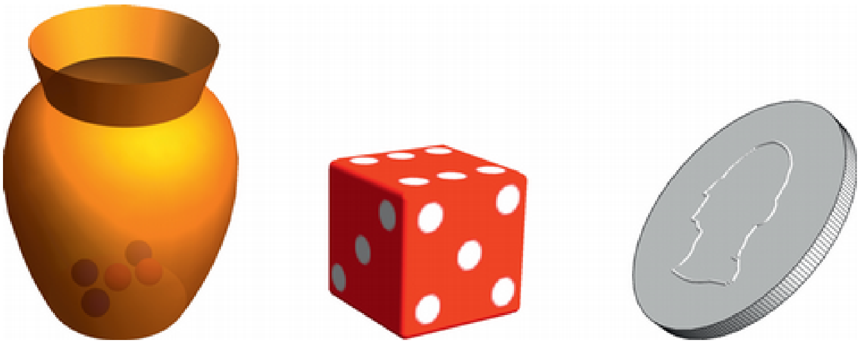

Figure 1.1. Three traditional mechanisms for creating experiments with random outcomes: an urn with balls, a six-sided die, and a coin.

Sampling with replacement, order matters

Suppose the urn contains n balls numbered $1, 2, \ldots, n$. We retrieve a ball from the urn, record its number, and put the ball back into the urn. (Putting the ball back into the urn is the *replacement* step.) We carry out this procedure k times. The outcome is the ordered k-tuple of numbers that we read off the sampled balls. Represent the outcome as $\omega = (s_1, s_2, \ldots, s_k)$ where s_1 is the number on the first ball, s_2 is the number on the second ball, and so on. The sample space Ω is a Cartesian product space: if we let $S = \{1, 2, \ldots, n\}$ then

$$\Omega = \underbrace{S \times S \times \cdots \times S}_{k \text{ times}} = S^k = \{(s_1, s_2, \ldots, s_k) : s_i \in S \text{ for } i = 1, \ldots, k\}. \tag{1.6}$$

How many outcomes are there? Each s_i can be chosen in n different ways. By Fact C.5 from Appendix C we have

$$\#\Omega = \underbrace{n \cdot n \cdots n}_{k \text{ times}} = n^k.$$

We assume that this procedure leads to equally likely outcomes, hence the probability of each k-tuple is $P\{\omega\} = n^{-k}$.

Let us illustrate this with a numerical example.

Example 1.10. Suppose our urn contains 5 balls labeled 1, 2, 3, 4, 5. Sample 3 balls with replacement and produce an ordered list of the numbers drawn. At each step we have the same 5 choices. The sample space is

$$\Omega = \{1, 2, 3, 4, 5\}^3 = \{(s_1, s_2, s_3) : \text{each } s_i \in \{1, 2, 3, 4, 5\}\}$$

and $\#\Omega = 5^3$. Since all outcomes are equally likely, we have for example

$$P\{\text{the sample is } (2,1,5)\} = P\{\text{the sample is } (2,2,3)\} = 5^{-3} = \frac{1}{125}. \qquad \blacktriangle$$

Repeated flips of a coin or rolls of a die are also examples of sampling with replacement. In these cases we are sampling from the set $\{\mathtt{H}, \mathtt{T}\}$ or $\{1, 2, 3, 4, 5, 6\}$.

(Check that Examples 1.6 and 1.7 are consistent with the language of sampling that we just introduced.)

Sampling without replacement, order matters

Consider again the urn with n balls numbered $1, 2, \ldots, n$. We retrieve a ball from the urn, record its number, and put the ball aside, in other words not back into the urn. (This is the *without replacement* feature.) We repeat this procedure k times. Again we produce an ordered k-tuple of numbers $\omega = (s_1, s_2, \ldots, s_k)$ where each $s_i \in S = \{1, 2, \ldots, n\}$. However, the numbers $s_1, s_2, \ldots, s_k$ in the outcome are distinct because now the same ball cannot be drawn twice. Because of this, we clearly cannot have k larger than n.

Our sample space is

$$\Omega = \{(s_1, s_2, \ldots, s_k) : \text{each } s_i \in S \text{ and } s_i \neq s_j \text{ if } i \neq j\}. \tag{1.7}$$

To find $\#\Omega$, note that s_1 can be chosen in n ways, after that s_2 can be chosen in $n-1$ ways, and so on, until there are $n-k+1$ choices remaining for the last entry s_k. Thus

$$\#\Omega = n \cdot (n-1) \cdot (n-2) \cdots (n-k+1) = (n)_k. \tag{1.8}$$

Again we assume that this mechanism gives us equally likely outcomes, and so $P\{\omega\} = \frac{1}{(n)_k}$ for each k-tuple ω of distinct numbers. The last symbol $(n)_k$ of equation (1.8) is called the descending factorial.

Example 1.11. Consider again the urn with 5 balls labeled 1, 2, 3, 4, 5. Sample 3 balls without replacement and produce an ordered list of the numbers drawn. Now the sample space is

$$\Omega = \{(s_1, s_2, s_3) : \text{each } s_i \in \{1, 2, 3, 4, 5\} \text{ and } s_1, s_2, s_3 \text{ are all distinct}\}.$$

The first ball can be chosen in 5 ways, the second ball in 4 ways, and the third ball in 3 ways. So

$$P\{\text{the sample is } (2,1,5)\} = \frac{1}{5 \cdot 4 \cdot 3} = \frac{1}{60}.$$

The outcome $(2, 2, 3)$ is not possible because repetition is not allowed. ▲

Another instance of sampling without replacement would be a random choice of students from a class to fill specific roles in a school play, with at most one role per student.

If $k = n$ then our sample is a random ordering of all n objects. Equation (1.8) becomes $\#\Omega = n!$. This is a restatement of the familiar fact that a set of n elements can be ordered in $n!$ different ways.

Sampling without replacement, order irrelevant

In the previous sampling situations the order of the outcome was relevant. That is, outcomes $(1, 2, 5)$ and $(2, 1, 5)$ were regarded as distinct. Next we suppose that we do not care about order, but only about the set $\{1, 2, 5\}$ of elements sampled. This kind of sampling without replacement can happen when cards are dealt from a deck or when winning numbers are drawn in a state lottery. Since order does not matter, we can also imagine choosing the entire set of k objects at once instead of one element at a time.

Notation is important here. The ordered triple $(1, 2, 5)$ and the set $\{1, 2, 5\}$ must not be confused with each other. Consequently *in this context* we must not mix up the notations ( ) and { }.

As above, imagine the urn with n balls numbered $1, 2, \ldots, n$. Let $1 \le k \le n$. Sample k balls without replacement, but record only which balls appeared and not the order. Since the sample contains no repetitions, the outcome is a subset of size k from the set $S = \{1, 2, \ldots, n\}$. Thus

$$\Omega = \{\omega \subset S : \#\omega = k\}.$$

(Do not be confused by the fact that an outcome ω is itself now a set of numbers.) The number of elements of Ω is given by the binomial coefficient (see Fact C.12 in Appendix C):

$$\#\Omega = \frac{n!}{(n-k)!k!} = \binom{n}{k}.$$

Assuming that the mechanism leads to equally likely outcomes, $P\{\omega\} = \binom{n}{k}^{-1}$ for each subset ω of size k.

Another way to produce an unordered sample of k balls without repetitions would be to execute the following three steps: (i) randomly order all n balls, (ii) take the first k balls, and (iii) ignore their order. Let us verify that the probability of obtaining a particular selection $\{s_1, \ldots, s_k\}$ is $\binom{n}{k}^{-1}$, as above.

The number of possible orderings in step (i) is $n!$. The number of favorable orderings is $k!(n-k)!$, because the first k numbers must be an ordering of $\{s_1, \ldots, s_k\}$ and after that comes an ordering of the remaining $n - k$ numbers. Then from the ratio of favorable to all outcomes

$$P\{\text{the selection is } \{s_1, \ldots, s_k\}\} = \frac{k!(n-k)!}{n!} = \frac{1}{\binom{n}{k}},$$

as we expected.

The description above contains a couple of lessons.

(i) There can be more than one way to build a probability model to solve a given problem. But a warning is in order: once an approach has been chosen, it must be followed consistently. Mixing up different representations will surely lead to an incorrect answer.

(ii) It may pay to introduce additional structure into the problem. The second approach introduced order into the calculation even though in the end we wanted an outcome without order.

Example 1.12. Suppose our urn contains 5 balls labeled 1, 2, 3, 4, 5. Sample 3 balls without replacement and produce an unordered set of 3 numbers as the outcome. The sample space is

$$\Omega = \{\omega : \omega \text{ is a 3-element subset of } \{1, 2, 3, 4, 5\}\}.$$

For example

$$P(\text{the sample is } \{1,2,5\}) = \frac{1}{\binom{5}{3}} = \frac{2!3!}{5!} = \frac{1}{10}.$$

The outcome $\{2, 2, 3\}$ does not make sense as a set of three numbers because of the repetition. ▲

The fourth alternative, sampling with replacement to produce an unordered sample, does not lead to equally likely outcomes. This scenario will appear naturally in Example 6.7 in Chapter 6.

Further examples

The next example contrasts all three sampling mechanisms.

Example 1.13. Suppose we have a class of 24 children. We consider three different scenarios that each involve choosing three children.

(a) Every day a random student is chosen to lead the class to lunch, without regard to previous choices. What is the probability that Cassidy was chosen on Monday and Wednesday, and Aaron on Tuesday?

This is sampling with replacement to produce an ordered sample. Over a period of three days the total number of different choices is 24^3. Thus

$$P\{(\text{Cassidy, Aaron, Cassidy})\} = 24^{-3} = \frac{1}{13{,}824}.$$

(b) Three students are chosen randomly to be class president, vice president, and treasurer. No student can hold more than one office. What is the probability that Mary is president, Cory is vice president, and Matt treasurer?

Imagine that we first choose the president, then the vice president, and then the treasurer. This is sampling without replacement to produce an ordered sample. Thus

$$P\{\text{Mary is president, Cory is vice president, and Matt treasurer}\}$$
$$= \frac{1}{24 \cdot 23 \cdot 22} = \frac{1}{12{,}144}.$$

Suppose we asked instead for the probability that Ben is either president or vice president. We apply formula (1.5). The number of outcomes in which Ben ends up as president is $1 \cdot 23 \cdot 22$ (1 choice for president, then 23 choices for vice president, and finally 22 choices for treasurer). Similarly the number of ways in which Ben ends up as vice president is $23 \cdot 1 \cdot 22$. So

$$P\{\text{Ben is president or vice president}\} = \frac{1 \cdot 23 \cdot 22 + 23 \cdot 1 \cdot 22}{24 \cdot 23 \cdot 22} = \frac{1}{12}.$$

(c) A team of three children is chosen at random. What is the probability that the team consists of Shane, Heather and Laura?

A team means here simply a set of three students. Thus we are sampling without replacement to produce a sample without order.

$$P(\text{the team is \{Shane, Heather, Laura\}}) = \frac{1}{\binom{24}{3}} = \frac{1}{2024}.$$

What is the probability that Mary is on the team? There are $\binom{23}{2}$ teams that include Mary since there are that many ways to choose the other two team members from the remaining 23 students. Thus by the ratio of favorable outcomes to all outcomes,

$$P\{\text{the team includes Mary}\} = \frac{\binom{23}{2}}{\binom{24}{3}} = \frac{3}{24} = \frac{1}{8}.$$ ▲

Problems of unordered sampling without replacement can be solved either with or without order. The next two examples illustrate this idea.

Example 1.14. Our urn contains 10 marbles numbered 1 to 10. We sample 2 marbles without replacement. What is the probability that our sample contains the marble labeled 1? Let A be the event that this happens. However we choose to count, the final answer $P(A)$ will come from formula (1.5).

Solution with order. Sample the 2 marbles in order. As in (1.8), $\#\Omega = 10\cdot 9 = 90$. The favorable outcomes are all the ordered pairs that contain 1:

$$A = \{(1, 2), (1, 3), \dots, (1, 10), (2, 1), (3, 1), \dots, (10, 1)\}$$

and we count $\#A = 18$. Thus $P(A) = \frac{18}{90} = \frac{1}{5}$.

Solution without order. Now the outcomes are subsets of size 2 from the set $\{1, 2, \dots, 10\}$ and so $\#\Omega = \binom{10}{2} = \frac{9\cdot 10}{2} = 45$. The favorable outcomes are all the 2-element subsets that contain 1:

$$A = \{\{1, 2\}, \{1, 3\}, \dots, \{1, 10\}\}.$$

Now $\#A = 9$ so $P(A) = \frac{9}{45} = \frac{1}{5}$.

Both approaches are correct and of course they give the same answer. ▲

Example 1.15. Rodney packs 3 shirts for a trip. It is early morning so he just grabs 3 shirts randomly from his closet. The closet contains 10 shirts: 5 striped, 3 plaid, and 2 solid colored ones. What is the probability that he chose 2 striped and 1 plaid shirt?

To use the counting methods introduced above, the shirts need to be distinguished from each other. This way the outcomes are equally likely. So let us assume that the shirts are labeled, with the striped shirts labeled 1, 2, 3, 4, 5, the plaid ones 6, 7, 8, and the solid colored ones 9, 10. Since we are only interested in the set of chosen shirts, we can solve this problem with or without order.

If we solve the problem without considering order, then

$$\Omega = \{\{x_1, x_2, x_3\} : x_i \in \{1, \dots, 10\}, x_i \neq x_j\},$$

the collection of 3-element subsets of $\{1, \dots, 10\}$. $\#\Omega = \binom{10}{3} = \frac{8 \cdot 9 \cdot 10}{2 \cdot 3} = 120$, the number of ways of choosing a set of 3 objects from a set of 10 objects. The set of favorable outcomes is

$$A = \{\{x_1, x_2, x_3\} : x_1, x_2 \in \{1, \dots, 5\}, x_1 \neq x_2, x_3 \in \{6, 7, 8\}\}.$$

The number of favorable outcomes is $\#A = \binom{5}{2} \cdot \binom{3}{1} = 30$. This comes from the number of ways of choosing 2 out of 5 striped shirts (shirts labeled 1, 2, 3, 4, 5) times the number of ways of choosing 1 out of 3 plaid shirts (numbered 6, 7, 8). Consequently

$$P\{2 \text{ striped and } 1 \text{ plaid}\} = \frac{\#A}{\#\Omega} = \frac{\binom{5}{2}\binom{3}{1}}{\binom{10}{3}} = \frac{30}{120} = \frac{1}{4}.$$

We now change perspective and solve the problem with an ordered sample. To avoid confusion, we denote our sample space by $\widetilde{\Omega}$:

$$\widetilde{\Omega} = \{(x_1, x_2, x_3) : x_i \in \{1, \dots 10\}, x_1, x_2, x_3 \text{ distinct}\}.$$

We have $\#\widetilde{\Omega} = 10 \cdot 9 \cdot 8 = 720$. The event of interest, $\widetilde{A}$, consists of those triples that have two striped shirts and one plaid shirt.

The elements of $\widetilde{A}$ can be found using the following procedure: (i) choose the plaid shirt (3 choices), (ii) choose the position of the plaid shirt in the ordering (3 choices), (iii) choose the first striped shirt and place it in the first available position (5 choices), (iv) choose the second striped shirt (4 choices). Thus, $\#A = 3 \cdot 3 \cdot 5 \cdot 4 = 180$, and $P(\widetilde{A}) = \frac{180}{720} = \frac{1}{4}$, which agrees with the answer above. ▲

1.3. Infinitely many outcomes

The next step in our development is to look at examples with infinitely many possible outcomes.

Example 1.16. Flip a fair coin until the first tails comes up. Record the number of flips required as the outcome of the experiment. What is the space Ω of possible outcomes? The number of flips needed can be any positive integer, hence Ω must contain all positive integers. We can also imagine the scenario where tails never comes up. This outcome is represented by ∞ (infinity). Thus

$$\Omega = \{\infty, 1, 2, 3, \ldots\}.$$

The outcome is k if and only if the first $k-1$ flips are heads and the kth flip is tails. As in Example 1.7, this is one of the 2^k equally likely outcomes when we flip a coin k times, so the probability of this event is 2^{-k}. Thus

$$P\{k\} = 2^{-k} \quad \text{for each positive integer } k. \tag{1.9}$$

It remains to figure out the probability $P\{\infty\}$. We can derive it from the axioms of probability:

$$1 = P(\Omega) = P\{\infty, 1, 2, 3, \ldots\} = P\{\infty\} + \sum_{k=1}^{\infty} P\{k\}.$$

From (1.9)

$$\sum_{k=1}^{\infty} P\{k\} = \sum_{k=1}^{\infty} 2^{-k} = 1, \tag{1.10}$$

which implies that $P\{\infty\} = 0$. ▲

Equation (1.9) defines the *geometric probability distribution* with success parameter 1/2 on the positive integers. On line (1.10) we summed up a geometric series. If you forgot how to do that, turn to Appendix D.

Notice that the example showed us something that agrees with our intuition, but is still quite nontrivial: the probability that we never see tails in repeated flips of a fair coin is zero. This phenomenon gets a different treatment in Example 1.22 below.

Example 1.17. We pick a real number uniformly at random from the closed unit interval $[0, 1]$. Let X denote the number chosen. "Uniformly at random" means that X is equally likely to lie anywhere in $[0, 1]$. Obviously $\Omega = [0, 1]$. What is the probability that X lies in a smaller interval $[a, b] \subseteq [0, 1]$? Since all locations for X are equally likely, it appears reasonable to stipulate that the probability that X is in $[a, b]$ should equal the proportion of $[0, 1]$ covered by $[a, b]$:

$$P\{X \text{ lies in the interval } [a, b]\} = b - a \quad \text{for } 0 \le a \le b \le 1. \tag{1.11}$$

Equation (1.11) defines the *uniform probability distribution* on the interval $[0, 1]$. We meet it again in Section 3.1 as part of a systematic treatment of probability distributions. ♣ ▲

Example 1.18. Consider a dartboard in the shape of a disk with a radius of 9 inches. The bullseye is a disk of diameter $\frac{1}{2}$ inch in the middle of the board. What is the probability that a dart randomly thrown on the board hits the bullseye? Let us assume that the dart hits the board at a uniformly chosen random location, that is, the dart is equally likely to hit anywhere on the board.

The sample space is a disk of radius 9. For simplicity take the center as the origin of our coordinate system, so

$$\Omega = \{(x, y) : x^2 + y^2 \leq 9^2\}.$$

Let A be the event that represents hitting the bullseye. This is the disk of radius $\frac{1}{4}$: $A = \{(x, y) : x^2 + y^2 \leq (1/4)^2\}$. The probability should be uniform on the disk Ω, so by analogy with the previous example,

$$P(A) = \frac{\text{area of } A}{\text{area of } \Omega} = \frac{\pi \cdot (1/4)^2}{\pi \cdot 9^2} = \frac{1}{36^2} \approx 0.00077. \qquad \blacktriangle$$

There is a significant difference between the sample space of Example 1.16 on the one hand, and the sample spaces of Examples 1.17 and 1.18 on the other. The set $\Omega = \{\infty, 1, 2, 3, \dots\}$ is *countably infinite* which means that its elements can be arranged in a sequence, or equivalently, labeled by positive integers. A countably infinite sample space works just like a finite sample space. To specify a probability measure P, it is enough to specify the probabilities of the outcomes and then derive the probability of each event by additivity:

$$P(A) = \sum_{\omega:\, \omega \in A} P\{\omega\} \quad \text{for any event } A \subset \Omega.$$

Finite and countably infinite sample spaces are both called *discrete sample spaces*.

By contrast, the unit interval, and any nontrivial subinterval of the real line, is *uncountable*. No integer labeling can cover all its elements. This is not trivial to prove and we shall not pursue it here. But we can see that it is impossible to define the probability measure of Example 1.17 by assigning probabilities to individual points. To argue this by contradiction, suppose some real number $x \in [0, 1]$ has positive probability $c = P\{x\} > 0$. Since all outcomes are equally likely in this example, it must be that $P\{x\} = c$ for all $x \in [0, 1]$. But this leads immediately to absurdity. If A is a set with k elements then $P(A) = kc$ which is greater than 1 if we take k large enough. The rules of probability have been violated. We must conclude that the probability of each individual point is zero:

$$P\{x\} = 0 \quad \text{for each} \quad x \in [0, 1]. \tag{1.12}$$

The consequence of the previous argument is that the definition of probabilities of events on an uncountable space must be based on something other than individual points. Examples 1.17 and 1.18 illustrate how to use *length* and *area* to model a uniformly distributed random point. Later we develop tools for building models where the random point is not uniform.

The issue raised here is also intimately tied with the additivity axiom (iii) of Definition 1.1. Note that the axiom requires additivity only for a *sequence* of pairwise disjoint events, and not for an uncountable collection of events. Example 1.17 illustrates this point. The interval $[0, 1]$ is the union of all the singletons $\{x\}$ over $x \in [0, 1]$. But $P([0, 1]) = 1$ while $P\{x\} = 0$ for each x. So there is no conceivable way in which the probability of $[0, 1]$ comes by adding together probabilities of points. Let us emphasize once more that this does not violate axiom (iii) because $[0, 1] = \bigcup_{x\in[0,1]}\{x\}$ is an uncountable union.

1.4. Consequences of the rules of probability

We record some consequences of the axioms in Definition 1.1 that are worth keeping in mind because they are helpful for calculations. The discussion relies on basic set operations reviewed in Appendix B.

Decomposing an event

The most obviously useful property of probabilities is the additivity property: if $A_1, A_2, A_3, \ldots$ are pairwise disjoint events and A is their union, then $P(A) = P(A_1) + P(A_2) + P(A_3) + \cdots$. Calculation of the probability of a complicated event A almost always involves decomposing A into smaller disjoint pieces whose probabilities are easier to find. The next two examples illustrate both finite and infinite decompositions.

Example 1.19. An urn contains 30 red, 20 green and 10 yellow balls. Draw two without replacement. What is the probability that the sample contains exactly one red or exactly one yellow? To clarify the question, it means the probability that the sample contains exactly one red, or exactly one yellow, *or both* (inclusive or). This interpretation of *or* is consistent with unions of events.

We approach the problem as we did in Example 1.15. We distinguish between the 60 balls for example by numbering them, though the actual labels on the balls are not important. This way we can consider an experiment with equally likely outcomes.

Having exactly one red or exactly one yellow ball in our sample of two means that we have one of the following color combinations: red-green, yellow-green or red-yellow. These are disjoint events, and their union is the event we are interested in. So

$$
\begin{aligned}
&P(\text{exactly one red or exactly one yellow}) \\
&\qquad = P(\text{red and green}) + P(\text{yellow and green}) + P(\text{red and yellow}).
\end{aligned}
$$

Counting favorable arrangements for each of the simpler events:

$$
P(\text{red and green}) = \frac{30 \cdot 20}{\binom{60}{2}} = \frac{20}{59}, \qquad P(\text{yellow and green}) = \frac{10 \cdot 20}{\binom{60}{2}} = \frac{20}{177},
$$

$$P(\text{red and yellow}) = \frac{30 \cdot 10}{\binom{60}{2}} = \frac{10}{59}.$$

This leads to

$$P(\text{exactly one red or exactly one yellow}) = \frac{20}{59} + \frac{20}{177} + \frac{10}{59} = \frac{110}{177}.$$

We used unordered samples, but we can get the answer also by using ordered samples. Example 1.24 below solves this same problem with inclusion-exclusion. ▲

Example 1.20. Peter and Mary take turns rolling a fair die. If Peter rolls 1 or 2 he wins and the game stops. If Mary rolls 3, 4, 5, or 6, she wins and the game stops. They keep rolling in turn until one of them wins. Suppose Peter rolls first.

(a) What is the probability that Peter wins and rolls at most 4 times?

To say that Peter wins and rolls at most 4 times is the same as saying that either he wins on his first roll, or he wins on his second roll, or he wins on his third roll, or he wins on his fourth roll. These alternatives are mutually exclusive. This is a fairly obvious way to decompose the event. So define events

$$A = \{\text{Peter wins and rolls at most 4 times}\}$$

and $A_k = \{\text{Peter wins on his } k\text{th roll}\}$. Then $A = \cup_{k=1}^{4} A_k$ and since the events A_k are mutually exclusive, $P(A) = \sum_{k=1}^{4} P(A_k)$.

To find the probabilities $P(A_k)$ we need to think about the game and the fact that Peter rolls first. Peter wins on his kth roll if first both Peter and Mary fail $k-1$ times and then Peter succeeds. Each roll has 6 possible outcomes. Peter's roll fails in 4 different ways and Mary's roll fails in 2 different ways. Peter's kth roll succeeds in 2 different ways. Thus the ratio of the number of favorable alternatives over the total number of alternatives gives

$$P(A_k) = \frac{(4 \cdot 2)^{k-1} \cdot 2}{(6 \cdot 6)^{k-1} \cdot 6} = \left(\frac{8}{36}\right)^{k-1} \frac{2}{6} = \left(\frac{2}{9}\right)^{k-1} \frac{1}{3}.$$

The probability asked is now obtained from a finite geometric sum:

$$\begin{aligned} P(A) &= \sum_{k=1}^{4} P(A_k) = \sum_{k=1}^{4} \left(\frac{2}{9}\right)^{k-1} \frac{1}{3} = \frac{1}{3} \sum_{j=0}^{3} \left(\frac{2}{9}\right)^{j} = \tfrac{1}{3} \cdot \frac{1-(\frac{2}{9})^4}{1-\frac{2}{9}} \\ &= \tfrac{3}{7}\left(1-(\tfrac{2}{9})^4\right). \end{aligned}$$

Above we changed the summation index to $j = k-1$ to make the sum look exactly like the one in equation (D.2) in Appendix D and then applied formula (D.2).

(b) What is the probability that Mary wins?

If Mary wins, then either she wins on her first roll, or she wins on her second roll, or she wins on her third roll, etc., and these alternatives are mutually exclusive. There is no a priori bound on how long the game can last. Hence we have to consider all the infinitely many possibilities.

Define the events $B = \{\text{Mary wins}\}$ and $B_k = \{\text{Mary wins on her } k\text{th roll}\}$. Then $B = \cup_{k=1}^{\infty} B_k$ is a union of pairwise disjoint events and the additivity of probability implies $P(B) = \sum_{k=1}^{\infty} P(B_k)$.

In order for Mary to win on her kth roll, first Peter and Mary both fail $k-1$ times, then Peter fails once more, and then Mary succeeds. Thus the ratio of the number of favorable alternatives over the total number of ways k rolls for both people can turn out gives

$$P(B_k) = \frac{(4 \cdot 2)^{k-1} \cdot 4 \cdot 4}{(6 \cdot 6)^k} = \left(\frac{8}{36}\right)^{k-1} \frac{16}{36} = \left(\frac{2}{9}\right)^{k-1} \frac{4}{9}.$$

The answer comes from a geometric series:

$$P\{\text{Mary wins}\} = P(B) = \sum_{k=1}^{\infty} P(B_k) = \sum_{k=1}^{\infty} \left(\frac{2}{9}\right)^{k-1} \frac{4}{9} = \frac{\frac{4}{9}}{1 - \frac{2}{9}} = \frac{4}{7}.$$

Note that we calculated the winning probabilities without defining the sample space. This will be typical going forward. Once we have understood the general principles of building probability models, it is usually not necessary to define explicitly the sample space in order to do calculations. ▲

Events and complements

Events A and A^c are disjoint and together make up Ω, no matter what the event A happens to be. Consequently

$$P(A) + P(A^c) = 1. \tag{1.13}$$

This identity tells us that to find $P(A)$, we can compute either $P(A)$ or $P(A^c)$. Sometimes one is much easier to compute than the other.

Example 1.21. Roll a fair die 4 times. What is the probability that some number appears more than once? A moment's thought reveals that this event contains a complicated collection of possible arrangements. Any one of 1 through 6 can appear two, three or four times. But also two different numbers can appear twice, and we would have to be careful not to overcount. However, switching to the complement provides an easy way out. If we set

$$A = \{\text{some number appears more than once}\}$$

then

$$A^c = \{\text{all rolls are different}\}.$$

By counting the possibilities $P(A^c) = \frac{6 \cdot 5 \cdot 4 \cdot 3}{6^4} = \frac{5}{18}$ and consequently $P(A) = \frac{13}{18}$. ▲

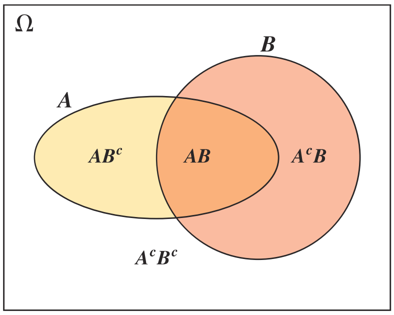

Figure 1.2. Venn diagram representation of two events A and B.

Equation (1.13) generalizes as follows. Intersecting with A and A^c splits any event B into two disjoint pieces $A \cap B$ and $A^c \cap B$, and so

$$P(B) = P(A \cap B) + P(A^c \cap B). \tag{1.14}$$

This identity is ubiquitous. Even in this section it appears several times.

There is an alternative way to write an intersection of sets: instead of $A \cap B$ we can write simply AB. Both will be used in the sequel. With this notation (1.14) is written as $P(B) = P(AB) + P(A^cB)$. The Venn diagram in Figure 1.2 shows a graphical representation of this identity.

Monotonicity of probability

Another intuitive and very useful fact is that a larger event must have larger probability:

$$\text{if } A \subseteq B \text{ then } P(A) \leq P(B). \tag{1.15}$$

If $A \subseteq B$ then $B = A \cup A^cB$ where the two events on the right are disjoint. (Figure 1.3 shows a graphical representation of this identity.) Now inequality (1.15) follows from the additivity and nonnegativity of probabilities:

$$P(B) = P(A) + P(B \cap A^c) \geq P(A).$$

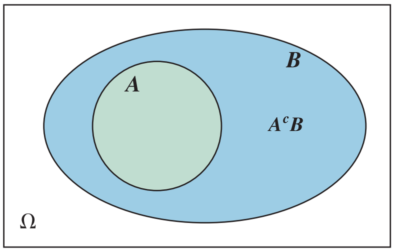

Figure 1.3. Venn diagram representation of the events $A \subseteq B$.

Example 1.22. Here is another proof of the fact, first seen in Example 1.16, that with probability 1 repeated flips of a fair coin eventually yield tails. Let A be the event that we never see tails and A_n the event that the first n coin flips are all heads. Never seeing tails implies that the first n flips must be heads, so $A \subseteq A_n$

and thus $P(A) \le P(A_n)$. Now $P(A_n) = 2^{-n}$ so we conclude that $P(A) \le 2^{-n}$. This is true for every positive n. The only nonnegative number $P(A)$ that can satisfy all these inequalities is zero. Consequently $P(A) = 0$.

The logic in this example is important. We can imagine an infinite sequence of flips all coming out heads. So this is not a logically impossible scenario. What we can say with mathematical certainty is that the heads-forever scenario has probability zero.

▲

Inclusion-exclusion

We move to inclusion-exclusion rules that tell us how to compute the probability of a union when the events are not mutually exclusive.

Fact 1.23. (Inclusion-exclusion formulas for two and three events)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B). \tag{1.16}$$

$$\begin{aligned} P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) \\ - P(B \cap C) + P(A \cap B \cap C). \end{aligned} \tag{1.17}$$

We prove identity (1.16). Look at the Venn diagram in Figure 1.2 to understand the first and third step below:

$$\begin{aligned} P(A \cup B) &= P(AB^c) + P(AB) + P(A^cB) \\ &= \big(P(AB^c) + P(AB)\big) + \big(P(AB) + P(A^cB)\big) - P(AB) \\ &= P(A) + P(B) - P(AB). \end{aligned}$$

Identity (1.17) can be proved similarly with a Venn diagram for three events.

Note that (1.16) can be rearranged to express the probability of the intersection from the individual probabilities and the probability of the union:

$$P(A \cap B) = P(A) + P(B) - P(A \cup B). \tag{1.18}$$

Example 1.24. (Example 1.19 revisited) An urn contains 30 red, 20 green and 10 yellow balls. Draw two without replacement. What is the probability that the sample contains exactly one red or exactly one yellow?

We solved this problem in Example 1.19 by breaking up the event into smaller parts. Now apply first inclusion-exclusion (1.16) and then count favorable arrangements using unordered samples:

$$\begin{aligned} &P(\text{exactly one red or exactly one yellow}) \\ &\quad = P(\{\text{exactly one red}\} \cup \{\text{exactly one yellow}\}) \end{aligned}$$

$$\begin{aligned}&= P(\text{exactly one red}) + P(\text{exactly one yellow}) \\ &\quad - P(\text{exactly one red and exactly one yellow}) \\ &= \frac{30\cdot 30}{\binom{60}{2}} + \frac{10\cdot 50}{\binom{60}{2}} - \frac{30\cdot 10}{\binom{60}{2}} = \frac{110}{177},\end{aligned}$$

which gives the same probability we obtained in Example 1.19. ▲

Example 1.25. In a town 15% of the population is blond, 25% of the population has blue eyes and 2% of the population is blond with blue eyes. What is the probability that a randomly chosen individual from the town is not blond and does not have blue eyes? (We assume that each individual has the same probability to be chosen.)

In order to translate the information into the language of probability, we identify the sample space and relevant events. The sample space Ω is the entire population of the town. The important events or subsets of Ω are

$$\begin{aligned}A &= \{\text{blond members of the population}\}, \qquad \text{and} \\ B &= \{\text{blue-eyed members of the population}\}.\end{aligned}$$

The problem gives us the following information:

$$P(A) = 0.15, \qquad P(B) = 0.25, \qquad \text{and} \qquad P(AB) = 0.02. \tag{1.19}$$

Our goal is to compute the probability of A^cB^c. At this point we could forget the whole back story and work with the following problem: suppose that (1.19) holds, find $P(A^cB^c)$.

By de Morgan's law (equation (B.1) on page 382) $A^cB^c = (A \cup B)^c$. Thus from the inclusion-exclusion formula we get

$$\begin{aligned}P(A^cB^c) &= 1 - P(A \cup B) = 1 - (P(A) + P(B) - P(AB)) \\ &= 1 - (0.15 + 0.25 - 0.02) = 0.62.\end{aligned}$$

Another way to get the same result is by applying (1.18) for A^c and B^c to express $P(A^cB^c)$ the following way

$$P(A^cB^c) = P(A^c) + P(B^c) - P(A^c \cup B^c).$$

We can compute $P(A^c)$ and $P(B^c)$ from $P(A)$ and $P(B)$. By de Morgan's law $A^c \cup B^c = (AB)^c$, so $P(A^c \cup B^c) = 1 - P(AB)$. Now we have all the ingredients and

$$P(A^cB^c) = (1 - 0.15) + (1 - 0.25) - (1 - 0.02) = 0.62.$$ ▲

The general formula is for any collection of n events $A_1, A_2, \ldots, A_n$ on a sample space.

Fact 1.26. (General inclusion-exclusion formula)

$$
\begin{aligned}
P(A_1 \cup \cdots \cup A_n) = &\sum_{i=1}^{n} P(A_i) - \sum_{1 \le i_1 < i_2 \le n} P(A_{i_1} \cap A_{i_2}) \\
&+ \sum_{1 \le i_1 < i_2 < i_3 \le n} P(A_{i_1} \cap A_{i_2} \cap A_{i_3}) \\
&- \sum_{1 \le i_1 < i_2 < i_3 < i_4 \le n} P(A_{i_1} \cap A_{i_2} \cap A_{i_3} \cap A_{i_4}) \\
&+ \cdots + (-1)^{n+1} P(A_1 \cap \cdots \cap A_n) \\
= &\sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \le i_1 < \cdots < i_k \le n} P(A_{i_1} \cap \cdots \cap A_{i_k}).
\end{aligned} \tag{1.20}
$$

Note that summing over $1 \le i_1 < i_2 < i_3 \le n$ is exactly the same as summing over all triples of distinct indices from $\{1, \dots, n\}$. The number of such triples is $\binom{n}{3}$. The last line of equation (1.20) is a succinct way of expressing the entire formula: the first index k specifies the number of events put in each intersection, and then indices $i_1 < \cdots < i_k$ pick the k events out of $A_1, \dots, A_n$ that are intersected in each term of the last sum.

Our last example is a probability classic.

Example 1.27. Suppose n people arrive for a show and leave their hats in the cloakroom. Unfortunately, the cloakroom attendant mixes up the hats completely so that each person leaves with a random hat. Let us assume that all $n!$ assignments of hats are equally likely. What is the probability that no one gets his/her own hat? How does this probability behave as $n \to \infty$?

Define the events

$$A_i = \{\text{person } i \text{ gets his/her own hat}\}, \qquad 1 \le i \le n.$$

The probability we want is

$$P\left(\bigcap_{i=1}^{n} A_i^c\right) = 1 - P\left(\bigcup_{i=1}^{n} A_i\right), \tag{1.21}$$

where we used de Morgan's law. We compute $P(A_1 \cup \cdots \cup A_n)$ with the inclusion-exclusion formula (1.20). Let $i_1 < i_2 < \cdots < i_k$ and evaluate

$$
\begin{aligned}
P(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}) &= P\{\text{individuals } i_1, i_2, \dots, i_k \text{ get their own hats}\} \\
&= \frac{(n-k)!}{n!},
\end{aligned} \tag{1.22}
$$

where the formula comes from counting favorable arrangements: if we assign k individuals their correct hats, there are $(n-k)!$ ways to distribute the remaining hats. (Note that the event $A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}$ does not say that these k are the *only* people who receive correct hats.) Thus

$$\sum_{i_1 < i_2 < \cdots < i_k} P(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}) = \binom{n}{k} \frac{(n-k)!}{n!} = \frac{1}{k!},$$

since there are $\binom{n}{k}$ terms in the sum. From (1.20)

$$P\left(\bigcup_{i=1}^{n} A_i\right) = 1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \cdots + (-1)^{n+1} \frac{1}{n!}$$

and then by (1.21)

$$P\left(\bigcap_{i=1}^{n} A_i^c\right) = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!} = \sum_{k=0}^{n} \frac{(-1)^k}{k!}. \tag{1.23}$$

This is the beginning of the familiar series representation of the function e^x at $x = -1$. (See (D.3) in Appendix D for a reminder.) Thus the limit as $n \to \infty$ is immediate:

$$\lim_{n\to\infty} P(\text{no person among } n \text{ people gets the correct hat}) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} = e^{-1}.$$

The reader with some background in algebra or combinatorics may notice that this example is really about the number of fixed points of a *random permutation*. A permutation of a set B is a bijective function $f : B \to B$. The fixed points of a permutation f are those elements x that satisfy $f(x) = x$. If we imagine that both the persons and the hats are numbered from 1 to n (with hat i belonging to person i) then we get a permutation that maps each person (or rather, her number) to the hat (or rather, its number) she receives. The result we derived says that as $n \to \infty$, the probability that a random permutation of n elements has no fixed points converges to e^{-1}. ▲

1.5. Random variables: a first look

In addition to the basic outcomes themselves, we are often interested in various numerical values derived from the outcomes. For example, in the game of Monopoly we roll a pair of dice, and the interesting outcome is the *sum* of the values of the dice. Or in a finite sequence of coin flips we might be interested in the total number of tails instead of the actual sequence of coin flips. This idea of attaching a number to each outcome is captured by the notion of a random variable.

Definition 1.28. Let Ω be a sample space. A **random variable** is a function from Ω into the real numbers. ♣

There are some conventions to get used to here. First the terminology: a random variable is not a variable but a function. Another novelty is the notation. In calculus we typically denote functions with lower case letters such as f, g and h. By contrast, random variables are usually denoted by capital letters such as X, Y and Z. The value of a random variable X at sample point ω is $X(\omega)$.

The study of random variables occupies much of this book. At this point we want to get comfortable with describing events in terms of random variables.

Example 1.29. We consider again the roll of a pair of dice (Example 1.6). Let us introduce three random variables: X_1 is the outcome of the first die, X_2 is the outcome of the second die, and S is the sum of the two dice. The precise definitions are these. For each sample point $(i,j) \in \Omega$,

$$X_1(i,j) = i, \quad X_2(i,j) = j, \quad \text{and} \quad S(i,j) = X_1(i,j) + X_2(i,j) = i + j.$$

To take a particular sample point, suppose the first die is a five and the second die is a one. Then the realization of the experiment is (5, 1) and the random variables take on the values

$$X_1(5,1) = 5, \quad X_2(5,1) = 1, \quad \text{and} \quad S(5,1) = X_1(5,1) + X_2(5,1) = 6.$$

In Example 1.6 we considered the event $D = \{\text{the sum of the two dice is } 8\}$ and computed its probability. In the language of random variables, $D = \{S = 8\}$ and

$$P\{S = 8\} = \sum_{(i,j):i+j=8} P\{X_1 = i,\ X_2 = j\} = \tfrac{5}{36}. \tag{1.24}$$

▲

A few more notational comments are in order. Recall that events are subsets of Ω. We write $\{S = 8\}$ for the set of sample points (i,j) such that $S(i,j) = 8$. The conventional full-fledged set notation for this is

$$\{(i,j) \in \Omega : S(i,j) = 8\}. \tag{1.25}$$

What we do in probability is drop all but the essential features of this notation. In most circumstances the expression $\{S = 8\}$ tells us all we need.

Inside events a comma is short for "and" which represents intersection. So the event on line (1.24) means the following:

$$\{X_1 = i,\ X_2 = j\} = \{X_1 = i \text{ and } X_2 = j\} = \{X_1 = i\} \cap \{X_2 = j\}.$$

If we wanted a union we would write "or":

$$\{X_1 = i \text{ or } X_2 = j\} = \{X_1 = i\} \cup \{X_2 = j\}.$$

Example 1.30. A die is rolled. If the outcome of the roll is 1, 2, or 3, the player loses \$1. If the outcome is 4, the player gains \$1, and if the outcome is 5 or 6, the player gains \$3. Let W denote the change in wealth of the player in one round of this game.

The sample space for the roll of the die is $\Omega = \{1, 2, 3, 4, 5, 6\}$. The random variable W is the real-valued function on Ω defined by

$$W(1) = W(2) = W(3) = -1, \quad W(4) = 1, \quad W(5) = W(6) = 3.$$

The probabilities of the values of W are

$$\begin{aligned} &P\{W = -1\} = P\{1, 2, 3\} = \tfrac{1}{2}, \quad P\{W = 1\} = P\{4\} = \tfrac{1}{6}, \\ &\text{and} \quad P\{W = 3\} = P\{5, 6\} = \tfrac{1}{3}. \end{aligned} \tag{1.26}$$

▲

Example 1.31. As in Example 1.17, select a point uniformly at random from the interval $[0, 1]$. Let Y be equal to twice the chosen point. The sample space is $\Omega = [0, 1]$, and the random variable is $Y(\omega) = 2\omega$. Let us compute $P\{Y \le a\}$ for $a \in [0, 2]$. By our convention discussed around (1.25), $\{Y \le a\} = \{\omega : Y(\omega) \le a\}$. Therefore

$$\{Y \le a\} = \{\omega : Y(\omega) \le a\} = \{\omega : 2\omega \le a\} = \{\omega : \omega \le a/2\} = [0, a/2],$$

where the second equality follows from the definition of Y, the third follows by algebra and the last is true because in this example the sample points ω are points in $[0, 1]$. Thus

$$P\{Y \le a\} = P([0, a/2]) = a/2 \qquad \text{for } a \in [0, 2]$$

by (1.11). If $a < 0$ then the event $\{Y \le a\}$ is empty, and consequently $P\{Y \le a\} = P(\varnothing) = 0$ for $a < 0$. If $a > 2$ then $P\{Y \le a\} = P(\Omega) = 1$. ▲

Example 1.32. A random variable X is *degenerate* if there is some real value b such that $P(X = b) = 1$. ▲

A degenerate random variable is in a sense not random at all because with probability 1 it has only one possible value. But it is an important special case to keep in mind. A real-valued function X on Ω is a *constant function* if there is some real value b such that $X(\omega) = b$ for all $\omega \in \Omega$. A constant function is a degenerate random variable. But a degenerate random variable does not have to be a constant function on all of Ω. Exercise 1.53 asks you to create an example.

As seen in all the examples above, events involving a random variable are of the form "the random variable takes certain values." The completely general form of such an event is

$$\{X \in B\} = \{\omega \in \Omega : X(\omega) \in B\}$$

where B is some subset of the real numbers. This reads "X lies in B."

Definition 1.33. Let X be a random variable. The **probability distribution** of the random variable X is the collection of probabilities $P\{X \in B\}$ for sets B of real numbers. ♣

The probability distribution of a random variable is an assignment of probabilities to subsets of $\mathbb{R}$ that satisfies again the axioms of probability (Exercise 1.54).

The next definition identifies a major class of random variables for which many exact computations are possible.

Definition 1.34. A random variable X is a **discrete random variable** if there exists a finite or countably infinite set $\{k_1, k_2, k_3, \dots\}$ of real numbers such that

$$\sum_i P(X = k_i) = 1 \tag{1.27}$$

where the sum ranges over the entire set of points $\{k_1, k_2, k_3, \dots\}$.

In particular, if the range of the random variable X is finite or countably infinite, then X is a discrete random variable. We say that those k for which $P(X = k) > 0$ are the *possible values* of the discrete random variable X.

In Example 1.29 the random variable S is the sum of two dice. The range of S is $\{2, 3, \dots, 12\}$ and so S is discrete.

In Example 1.31 the random variable Y is defined as twice a uniform random number from $[0, 1]$. For each real value a, $P(Y = a) = 0$ (see (1.12)), and so Y is *not* a discrete random variable.

The probability distribution of a discrete random variable is described completely in terms of its probability mass function.

Definition 1.35. The **probability mass function** (p.m.f.) of a discrete random variable X is the function p (or p_X) defined by

$$p(k) = P(X = k)$$

for possible values k of X.

The function p_X gives the probability of each possible value of X. Probabilities of other events of X then come by additivity: for any subset $B \subseteq \mathbb{R}$,

$$P(X \in B) = \sum_{k \in B} P(X = k) = \sum_{k \in B} p_X(k), \tag{1.28}$$

where the sum is over the possible values k of X that lie in B. A restatement of equation (1.27) gives $\sum_k p_X(k) = 1$ where the sum extends over all possible values k of X.

Example 1.36. (Continuation of Example 1.29) Here are the probability mass functions of the first die and the sum of the dice.

k	1	2	3	4	5	6
$p_{X_1}(k) = P(X_1 = k)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

k	2	3	4	5	6	7	8	9	10	11	12
$p_S(k) = P(S = k)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Probabilities of events are obtained by summing values of the probability mass function. For example,

$$P(2 \le S \le 5) = p_S(2) + p_S(3) + p_S(4) + p_S(5) = \tfrac{1}{36} + \tfrac{2}{36} + \tfrac{3}{36} + \tfrac{4}{36} = \tfrac{10}{36}. \quad \blacktriangle$$

Example 1.37. (Continuation of Example 1.30) Here are the values of the probability mass function of the random variable W defined in Example 1.30: $p_W(-1) = \frac{1}{2}$, $p_W(1) = \frac{1}{6}$, and $p_W(3) = \frac{1}{3}$. ▲

We finish this section with an example where the probability mass function of a discrete random variable is calculated with the help of a random variable whose range is an interval.

Example 1.38. We have a dartboard of radius 9 inches. The board is divided into four parts by three concentric circles of radii 1, 3, and 6 inches. If our dart hits the smallest disk, we get 10 points, if it hits the next region then we get 5 points, and we get 2 and 1 points for the other two regions (see Figure 1.4). Let X denote

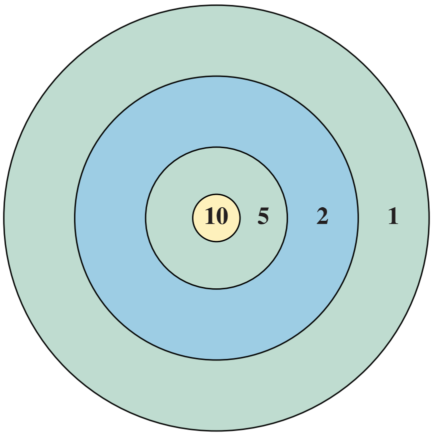

Figure 1.4. The dartboard for Example 1.38. The radii of the four circles in the picture are 1, 3, 6 and 9 inches.

the number of points we get when we throw a dart randomly (uniformly) at the board. How can we determine the distribution of X?

The possible values of X are $1, 2, 5$ and 10. We have to find the probabilities of these outcomes, that is, the probability mass function of X. It helps if we introduce another random variable: let $R \in [0, 9]$ denote the distance of the dart to the center. Then by Example 1.18 for any $a \in [0, 9]$ we have

$$P\{R \le a\} = \frac{\pi \cdot a^2}{\pi \cdot 9^2} = \frac{a^2}{81}. \tag{1.29}$$

From this we can compute the probability mass function of X:

$$\begin{aligned}
P\{X = 10\} &= P\{R \le 1\} = \frac{1}{81},\\
P\{X = 5\} &= P\{1 < R \le 3\} = P\{R \le 3\} - P\{R \le 1\} = \frac{8}{81},\\
P\{X = 2\} &= P\{3 < R \le 6\} = P\{R \le 6\} - P\{R \le 3\} = \frac{27}{81},\\
P\{X = 1\} &= P\{6 < R \le 9\} = P\{R \le 9\} - P\{R \le 6\} = \frac{45}{81}.
\end{aligned}$$

Note that the sum of the probabilities is equal to 1 (as it should be). ▲

1.6. Finer points ♣

Continuity of the probability measure

The following fact expresses a continuity property of probability measures as functions of events. See Exercise 1.61 for the companion statement for decreasing sequences.

Fact 1.39. Suppose we have an infinite sequence of events $A_1, A_2, A_3, \dots$ that are nested increasing: $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots \subseteq A_n \subseteq \cdots$. Let $A_\infty = \cup_{k=1}^{\infty} A_k$ denote the union. Then

$$\lim_{n\to\infty} P(A_n) = P(A_\infty). \tag{1.30}$$

Another way to state this is as follows: if we have increasing events then the probability of the union of all the events (the probability that at least one of them happens) is equal to the limit of the individual probabilities.

Recall from calculus that a function $f : \mathbb{R} \to \mathbb{R}$ is continuous at x if and only if for each sequence of points $x_1, x_2, x_3, \dots$ that converge to x, we have $f(x_n) \to f(x)$ as $n \to \infty$. In a natural way increasing sets A_n converge to their union A_∞, because the difference $A_\infty \setminus A_n$ shrinks away as $n \to \infty$. Thus (1.30) is an analogue of the calculus notion of continuity.

Proof of Fact 1.39. To take advantage of the additivity axiom of probability, we break up the events A_n into disjoint pieces. For $n = 2, 3, 4, \dots$ let $B_n = A_n \setminus A_{n-1}$.

Since each B_n is contained in A_n and disjoint from A_{n-1}, the events B_n are pairwise disjoint. We have the disjoint decomposition of A_n as

$$\begin{aligned} A_n &= A_{n-1} \cup B_n = A_{n-2} \cup B_{n-1} \cup B_n \\ &= \cdots = A_1 \cup B_2 \cup B_3 \cup \cdots \cup B_n. \end{aligned} \tag{1.31}$$

Taking a union of all the events A_n gives us the disjoint decomposition

$$\cup_{n=1}^{\infty} A_n = A_1 \cup B_2 \cup B_3 \cup B_4 \cup \cdots$$

By the additivity of probability, and by expressing the infinite series as the limit of partial sums,

$$\begin{aligned} P(\cup_{n=1}^{\infty} A_n) &= P(A_1) + P(B_2) + P(B_3) + (B_4) + \cdots \\ &= \lim_{n\to\infty} (P(A_1) + P(B_2) + \cdots + P(B_n)) = \lim_{n\to\infty} P(A_n). \end{aligned}$$

The last equality above came from the decomposition in (1.31). ▲

Measurability

The notion of an event is actually significantly deeper than we let on in this chapter. But for discrete sample spaces there is no difficulty. Usually each sample point ω has a well-defined probability and then each subset A of Ω has the probability given by the sum

$$P(A) = \sum_{\omega:\, \omega\in A} P\{\omega\}. \tag{1.32}$$

Thus every subset of a discrete sample space Ω is a legitimate event. For example, for the flip of a single coin the sample space is $\Omega = \{\mathtt{H}, \mathtt{T}\}$ and the collection of events is $\mathcal{F} = \{\varnothing, \{\mathtt{H}\}, \{\mathtt{T}\}, \{\mathtt{H}, \mathtt{T}\}\}$, which is exactly the collection of *all* subsets of Ω. The collection of all subsets of Ω is called the *power set* of Ω and is denoted by $2^{\Omega} = \{A : A \text{ is a subset of } \Omega\}$.

This all seems very straightforward. But it turns out that there can be good reasons to use smaller collections $\mathcal{F}$ of events. One reason is that a smaller $\mathcal{F}$ can be useful for modeling purposes. This is illustrated in Example 1.40 and Exercise 1.63 below. A second reason is that technical problems with uncountable sample spaces *prevent* us from taking $\mathcal{F}$ as the power set. We discuss this issue next.

Recall from Example 1.17 that for the uniform probability measure on $[0, 1]$ (1.32) fails for intervals because $P\{\omega\} = 0$ for each sample point ω. However, the technical issues go beyond this. It turns out that it is *impossible* to define consistently the uniform probability measure for all subsets of $[0, 1]$. Hence $\mathcal{F}$ must be something smaller than the power set of $[0, 1]$, but what exactly?

To put the theory on a sound footing, the axiomatic framework of Definition 1.1 is extended to impose the following requirements on the collection of events $\mathcal{F}$:

(i) the empty set $\varnothing$ is a member of $\mathcal{F}$,
(ii) if A is in $\mathcal{F}$, then A^c is also in $\mathcal{F}$,
(iii) if $A_1, A_2, A_3, \ldots$ is a sequence of events in $\mathcal{F}$, then their union $\cup_{i=1}^{\infty} A_i$ is also in $\mathcal{F}$.

Any collection $\mathcal{F}$ of sets satisfying properties (i)–(iii) is called a *σ-algebra* or a *σ-field*. The members of a σ-algebra are called *measurable sets*. The properties of a σ-algebra imply that countably many applications of the usual set operations to events is a safe way to produce new events.

To create the σ-algebra for the probability model we desire, we typically start with some nice sets that we want in $\mathcal{F}$, and let the axioms determine which other sets must also be members of $\mathcal{F}$. When Ω is $[0, 1]$, the real line, or some other interval of real numbers, the standard choice for $\mathcal{F}$ is to take the *smallest σ-algebra that contains all subintervals of* Ω. The members of this $\mathcal{F}$ are called *Borel sets*. It is impossible to give a simple description of all Borel sets. But this procedure allows us to rule out pathological sets for which probabilities cannot be assigned. Construction of probability spaces belongs in a subject called *measure theory* and lies beyond the scope of this book.

Similar issues arise with random variables. Not every function X from an uncountable Ω into $\mathbb{R}$ can be a random variable. We have to require that $\{X \in B\}$ is a member of $\mathcal{F}$ whenever B is a Borel subset of $\mathbb{R}$. This defines the notion of a *measurable function*, and it is the measurable functions that can appear as random variables. In particular, in Definition 1.33 the probability distribution $P\{X \in B\}$ of a random variable X is defined only for Borel sets $B \subseteq \mathbb{R}$.

Fortunately it turns out that all reasonable sets and functions encountered in practice are measurable. Consequently this side of the theory can be completely ignored in an undergraduate probability course. Readers who wish to pursue the full story can turn to graduate textbooks on real analysis and probability, such as [Bil95], [Fol99], and [Dur10].

Collections of events as information

Another aspect of the collection $\mathcal{F}$ of events is that it can represent *information*. This point becomes pronounced in advanced courses of probability. The next example gives a glimpse of the idea.

Example 1.40. (Continuation of Example 1.6) Consider again the experiment of rolling a pair of dice but this time we cannot tell the two dice apart. This means that we cannot separate outcomes $(1, 3)$ and $(3, 1)$ from each other. For a concrete experimental setup, imagine that I roll the dice behind a curtain and then tell you, "you got a one and a three," without specifying which die gave which number.

We can still model the situation with the same sample space $\Omega = \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\}$ of ordered pairs. By a judicious choice of $\mathcal{F}$ we can forbid the model from separating $(1, 3)$ and $(3, 1)$. This is how we define the collection of events: $\mathcal{F}$ consists of

- the singletons $\{(i, i)\}$ for integers $1 \leq i \leq 6$,
- the sets of pairs $\{(i, j), (j, i)\}$ for $1 \leq i, j \leq 6$ such that $i \neq j$,
- and all unions of sets of the above two types, including the empty set.

The probability measure P is the same as before, except that its domain is now the smaller collection $\mathcal{F}$ of events described above. $(1, 3)$ and $(3, 1)$ are still separate points of the space Ω, but $\{(1, 3)\}$ is not a member of $\mathcal{F}$ and hence does not have a probability assigned to it. The event $\{(1, 3), (3, 1)\}$ does have a probability assigned to it, namely $P\{(1, 3), (3, 1)\} = \frac{2}{36}$ as before. The point of this example is that by restricting $\mathcal{F}$ we can model the information available to the observer of the experiment, without changing Ω. ▲

Exercises

We start with some warm-up exercises arranged by section.

Section 1.1

Exercise 1.1. We roll a fair die twice. Describe a sample space Ω and a probability measure P to model this experiment. Let A be the event that the second roll is larger than the first. Find the probability $P(A)$ that the event A occurs.

Exercise 1.2. For breakfast Bob has three options: cereal, eggs or fruit. He has to choose exactly two items out of the three available.

(a) Describe the sample space of this experiment.

Hint. What are the different possible outcomes for Bob's breakfast?

(b) Let A be the event that Bob's breakfast includes cereal. Express A as a subset of the sample space.

Exercise 1.3.

(a) You flip a coin and roll a die. Describe the sample space of this experiment.
(b) Now each of 10 people flips a coin and rolls a die. Describe the sample space of this experiment. How many elements are in the sample space?
(c) In the experiment of part (b), how many outcomes are in the event where nobody rolled a five? How many outcomes are in the event where at least one person rolled a five?

Section 1.2

Exercise 1.4. Every day a kindergarten class chooses randomly one of the 50 state flags to hang on the wall, without regard to previous choices. We are interested in the flags that are chosen on Monday, Tuesday and Wednesday of next week.

(a) Describe a sample space Ω and a probability measure P to model this experiment.
(b) What is the probability that the class hangs Wisconsin's flag on Monday, Michigan's flag on Tuesday, and California's flag on Wednesday?
(c) What is the probability that Wisconsin's flag will be hung at least two of the three days?

Exercise 1.5. In one type of state lottery 5 distinct numbers are picked from $1, 2, 3, \ldots, 40$ uniformly at random.

(a) Describe a sample space Ω and a probability measure P to model this experiment.
(b) What is the probability that out of the five picked numbers exactly three will be even?

Exercise 1.6. We have an urn with 3 green and 4 yellow balls. We choose 2 balls randomly without replacement. Let A be the event that we have two different colored balls in our sample.

(a) Describe a possible sample space with equally likely outcomes, and the event A in your sample space.
(b) Compute $P(A)$.

Exercise 1.7. We have an urn with 3 green and 4 yellow balls. We draw 3 balls one by one without replacement.

(a) Find the probability that the colors we see in order are green, yellow, green.
(b) Find the probability that our sample of 3 balls contains 2 green balls and 1 yellow ball.

Exercise 1.8. Suppose that a bag of scrabble tiles contains 5 Es, 4 As, 3 Ns and 2 Bs. It is my turn and I draw 4 tiles from the bag without replacement. Assume that my draw is uniformly random. Let C be the event that I got two Es, one A and one N.

(a) Compute $P(C)$ by imagining that the tiles are drawn one by one as an ordered sample.
(b) Compute $P(C)$ by imagining that the tiles are drawn all at once as an unordered sample.

Section 1.3

Exercise 1.9. We break a stick at a uniformly chosen random location. Find the probability that the shorter piece is less than $1/5$th of the original.

Exercise 1.10. We roll a fair die repeatedly until we see the number four appear and then we stop. The outcome of the experiment is the number of rolls.

(a) Following Example 1.16 describe a sample space Ω and a probability measure P to model this situation.

(b) Calculate the probability that the number four never appears.

Exercise 1.11. We throw a dart at a square shaped board of side length 20 inches. Assume that the dart hits the board at a uniformly chosen random point. Find the probability that the dart is within 2 inches of the center of the board.

Section 1.4

Exercise 1.12. We roll a fair die repeatedly until we see the number four appear and then we stop.

(a) What is the probability that we need at most 3 rolls?
(b) What is the probability that we needed an even number of die rolls?

Exercise 1.13. At a certain school, 25% of the students wear a watch and 30% wear a bracelet. 60% of the students wear neither a watch nor a bracelet.

(a) One of the students is chosen at random. What is the probability that this student is wearing a watch or a bracelet?
(b) What is the probability that this student is wearing both a watch and a bracelet?

Exercise 1.14. Assume that $P(A) = 0.4$ and $P(B) = 0.7$. Making no further assumptions on A and B, show that $P(AB)$ satisfies $0.1 \leq P(AB) \leq 0.4$.

Exercise 1.15. An urn contains 4 balls: 1 white, 1 green and 2 red. We draw 3 balls with replacement. Find the probability that we did not see all three colors. Use two different calculations, as specified by (a) and (b) below.

(a) Define the event $W = \{\text{white ball did not appear}\}$ and similarly for G and R. Use inclusion-exclusion.
(b) Compute the probability by considering the complement of the event that we did not see all three colors.

Section 1.5

Exercise 1.16. We flip a fair coin five times. For every heads you pay me \$1 and for every tails I pay you \$1. Let X denote my net winnings at the end of five flips. Find the possible values and the probability mass function of X.

Exercise 1.17. An urn contains 4 red balls and 3 green balls. Two balls are drawn randomly.

(a) Let Z denote the number of green balls in the sample when the draws are done without replacement. Give the possible values and the probability mass function of Z.

(b) Let W denote the number of green balls in the sample when the draws are done with replacement. Give the possible values and the probability mass function of W.

Exercise 1.18. The statement

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SOME DOGS ARE BROWN
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has 16 letters. Choose one of the 16 letters uniformly at random. Let X denote the length of the *word* containing the chosen letter. Determine the possible values and probability mass function of X.

Exercise 1.19. You throw a dart and it lands uniformly at random on a circular dartboard of radius 6 inches. If your dart gets to within 2 inches of the center I will reward you with 5 dollars. But if your dart lands farther than 2 inches away from the center I will give you only 1 dollar. Let X denote the amount of your reward in dollars. Find the possible values and the probability mass function of X.

Further exercises

Exercise 1.20. We roll a fair die four times.

(a) Describe the sample space Ω and the probability measure P that model this experiment. To describe P, give the value $P\{\omega\}$ for each outcome $\omega \in \Omega$.
(b) Let A be the event that there are at least two fives among the four rolls. Let B be the event that there is at most one five among the four rolls. Find the probabilities $P(A)$ and $P(B)$ by finding the ratio of the number of favorable outcomes to the total, as in Fact 1.8.
(c) What is the set $A \cup B$? What equality should $P(A)$ and $P(B)$ satisfy? Check that your answers to part (b) satisfy this equality.

Exercise 1.21. Suppose an urn contains three black chips, two red chips, and two green chips. We draw three chips at random without replacement. Let A be the event that all three chips are of different color.

(a) Compute $P(A)$ by imagining that the chips are drawn one by one as an ordered sample.
(b) Compute $P(A)$ by imagining that the three chips are drawn all at once as an unordered sample.

Exercise 1.22. We pick a card uniformly at random from a standard deck of 52 cards. (If you are unfamiliar with the deck of 52 cards, see the description above Example C.19 in Appendix C.)

(a) Describe the sample space Ω and the probability measure P that model this experiment.

(b) Give an example of an event in this probability space with probability $\frac{3}{52}$.
(c) Show that there is no event in this probability space with probability $\frac{1}{5}$.

Exercise 1.23. The **Monty Hall problem** is a famous math problem loosely based on a television quiz show hosted by Monty Hall. You (the contestant) face three closed doors. There is a big prize behind one door, but nothing behind the other two. You are asked to pick one door out of the three, but you cannot yet see what is behind it. Monty opens one of the two other doors to reveal that there is nothing behind it. Then he offers you one chance to switch your choice. Should you switch?

This question has generated a great deal of confusion. An intuitively appealing quick answer goes like this. After Monty showed you a door with no prize, the prize is equally likely to be behind either one of the other two doors. Thus switching makes no difference. Is this correct?

To get clarity, let us use the simplest possible model. Let the outcome of the experiment be the door behind which the prize hides. So the sample space is $\Omega = \{1, 2, 3\}$. Assume that the prize is equally likely to be behind any one of the doors, and so we set $P\{1\} = P\{2\} = P\{3\} = 1/3$. Your decision to switch or not is not part of the model. Rather, use the model to compute the probability of winning separately for the two scenarios. We can fix the labeling of the doors so that your initial choice is door number 1. In both parts (a) and (b) below determine, for each door i, whether you win or not when the prize is behind door i.

(a) Suppose you will not switch. What is the probability that you win the prize?
(b) Suppose you *will* switch after Monty shows a door without a prize. What is the probability that you win the prize?

Exercise 1.24. Suppose the procedure of the quiz show is as described in the previous Exercise 1.23: you choose a door, Monty opens one of the two other doors with no prize, and you get to switch if you wish. Now suppose we have three positive numbers p_1, p_2, p_3 such that $p_1 + p_2 + p_3 = 1$ and the prize is behind door i with probability p_i. By labeling the doors suitably we can assume that $p_1 \geq p_2 \geq p_3$. Assume that you know the probabilities p_1, p_2, p_3 associated to each door. What is the strategy that maximizes your chances of winning the prize?

Exercise 1.25. A student lives within walking distance of 6 restaurants. Unbeknownst to her, she has exactly 1 friend at 3 of the restaurants, she has 2 friends at 2 of the restaurants, and she has no friends at one of the restaurants. Suppose that this accounts for all of her friends.

(a) Suppose that she chooses a restaurant at random. What is the probability that she has at least one friend at the restaurant?

(b) Suppose she calls one of her friends at random in order to meet up for dinner. What is the probability that she called someone at a restaurant where two of her friends are present?

Exercise 1.26. 10 men and 5 women are meeting in a conference room. Four people are chosen at random from the 15 to form a committee.

(a) What is the probability that the committee consists of 2 men and 2 women?
(b) Among the 15 is a couple, Bob and Jane. What is the probability that Bob and Jane both end up on the committee?
(c) What is the probability that Bob ends up on the committee but Jane does not?

Exercise 1.27. Suppose an urn contains seven chips labeled $1, \dots, 7$. Three of the chips are black, two are red, and two are green. The chips are drawn randomly one at a time without replacement until the urn is empty.

(a) What is the probability that the ith draw is chip number 5?
(b) What is the probability that the ith draw is black?

Exercise 1.28. We have an urn with m green balls and n yellow balls. Two balls are drawn at random. What is the probability that the two balls have the same color?

(a) Assume that the balls are sampled without replacement.
(b) Assume that the balls are sampled with replacement.
(c) When is the answer to part (b) larger than the answer to part (a)? Justify your answer. Can you give an intuitive explanation for what the calculation tells you?

Exercise 1.29. At a political meeting there are 7 liberals and 6 conservatives. We choose five people uniformly at random to form a committee. Let A be the event that we end up with a committee consisting of more conservatives than liberals.

(a) Describe a possible sample space for this experiment, and the event A in your sample space.
(b) Compute $P(A)$.

Exercise 1.30. Eight rooks are placed randomly on a chess board. What is the probability that none of the rooks can capture any of the other rooks? Translation for those who are not familiar with chess: pick 8 unit squares at random from an 8×8 square grid. What is the probability that no two chosen squares share a row or a column?

Hint. You can think of placing the rooks either with or without order, both approaches work.

Exercise 1.31. Two cards are dealt from an ordinary deck of 52 cards. This means that two cards are sampled uniformly at random without replacement.

(a) What is the probability that both cards are aces and one of them is the ace of spades?
(b) What is the probability that at least one of the cards is an ace?

Exercise 1.32. You are dealt five cards from a standard deck of 52. Find the probability of being dealt a *full house*. (This means that you have three cards of one face value and two cards of a different face value. An example would be three queens and two fours. See Exercise C.10 in Appendix C for a description of all the poker hands.)

Exercise 1.33. You roll a fair die 5 times. What is the probability of seeing a “full house” in the sense of seeing three rolls of one type, and two rolls of another different type? Note that we do not allow for all five rolls to be of the same type.

Exercise 1.34. Pick a uniformly chosen random point inside a unit square (a square of sidelength 1) and draw a circle of radius 1/3 around the point. Find the probability that the circle lies entirely inside the square.

Exercise 1.35. Pick a uniformly chosen random point inside the triangle with vertices $(0, 0)$, $(3, 0)$ and $(0, 3)$.

(a) What is the probability that the distance of this point to the y-axis is less than 1?
(b) What is the probability that the distance of this point to the origin is more than 1?

Exercise 1.36.

(a) Let (X, Y) denote a uniformly chosen random point inside the unit square

$$[0, 1]^2 = [0, 1] \times [0, 1] = \{(x, y) : 0 \le x, y \le 1\}.$$

Let $0 \le a < b \le 1$. Find the probability $P(a < X < b)$, that is, the probability that the x-coordinate X of the chosen point lies in the interval (a, b).
(b) What is the probability $P(|X - Y| \le 1/4)$?

Exercise 1.37. These questions pertain to Example 1.20 where Peter and Mary take turns rolling a fair die. To answer the questions, be precise about the definitions of your events and their probabilities.

(a) As in Example 1.20, suppose Peter takes the first roll. What is the probability that Mary wins and her last roll is a six?
(b) Suppose Mary takes the first roll. What is the probability that Mary wins?
(c) What is the probability that the game lasts an even number of rolls? Consider separately the case where Peter takes the first roll and the case

where Mary takes the first roll. Based on your intuition, which case should be likelier to end in an even number of rolls? Does the calculation confirm your intuition?

(d) What is the probability that the game does not go beyond k rolls? Consider separately the case where Peter takes the first roll and the case where Mary takes the first roll. Based on your intuition, in which case should the game be over faster? Does the calculation confirm your intuition?

Hint. Consider separately games that last $2j$ rolls (even number of rolls) and games that last $2j - 1$ rolls (odd number of rolls).

Exercise 1.38. Show that it is not possible to choose a uniform positive integer at random. (In other words, we cannot define a probability measure on the positive integers that can be considered uniform.)

Hint. What would be the probability of choosing a particular number?

Exercise 1.39. A restaurant offers three ways of paying the bill: cash, check, and card. At the end of the day there were 100 paid bills. Some of the bills were paid with a mix of cash, check, and card. There are 78 cash receipts, 26 card receipts and 16 checks. There were 13 bills that used cash and card, 4 that used card and check, 6 that used cash and check, and 3 that used all three methods of payment.

(a) If a bill is chosen at random, what is the probability that the bill was paid fully using only one of the methods?

(b) If two distinct bills are chosen at random, what is the probability that at least one was paid using two or more methods?

Exercise 1.40. An urn contains 1 green ball, 1 red ball, 1 yellow ball and 1 white ball. I draw 4 balls with replacement. What is the probability that there is at least one color that is repeated exactly twice?

Hint. Use inclusion-exclusion with events $G = \{\text{exactly two balls are green}\}$, $R = \{\text{exactly two balls are red}\}$, etc.

Exercise 1.41. Imagine a game of three players where exactly one player wins in the end and all players have equal chances of being the winner. The game is repeated four times. Find the probability that there is at least one person who wins no games.

Hint. Let $A_i = \{\text{person } i \text{ wins no games}\}$ and utilize the inclusion-exclusion formula.

Exercise 1.42. Suppose $P(A) > 0.8$ and $P(B) > 0.5$. Show that $P(AB) > 0.3$.

Exercise 1.43. Show that for any events $A_1, A_2, \ldots, A_n$,

$$P(A_1 \cup \cdots \cup A_n) \leq \sum_{k=1}^{n} P(A_k). \tag{1.33}$$

Hint. Obtain the case $n = 2$ from inclusion-exclusion, and more generally apply the $n = 2$ case repeatedly. Or formulate a proof by induction.

Exercise 1.44. Two fair dice are rolled. Let X be the maximum of the two numbers and Y the minimum of the two numbers on the dice.

(a) Find the possible values of X and the possible values of Y.

(b) Find the probabilities $P(X \leq k)$ for all integers k. Find the probability mass function of X.

Hint. Noticing that $P(X = k) = P(X \leq k) - P(X \leq k - 1)$ can save you some work.

(c) Find the probability mass function of Y.

Exercise 1.45. Imagine the following game of chance. There are four dollar bills on the table. You roll a fair die repeatedly. Every time you fail to get a six, one dollar bill is removed. When you get your first six, you get to take the money that remains on the table. If the money runs out before you get a six, you have lost and the game is over. Let X be the amount of your award. Find the possible values and the probability mass function of X.

Exercise 1.46. An urn contains 3 red balls and 1 yellow ball. We draw balls from the urn one by one without replacement until we see the yellow ball. Let X denote the number of red balls we see before the yellow. Find the possible values and the probability mass function of X.

Exercise 1.47. Consider the experiment of drawing a point uniformly at random from the unit interval, as in Examples 1.17 and 1.31, with $\Omega = [0, 1]$ and P determined by $P([a, b]) = b - a$ for $[a, b] \subseteq \Omega$. Define $Z(\omega) = e^{\omega}$ for $\omega \in \Omega$. Find $P(Z \leq t)$ for all real values t.

Exercise 1.48. Consider the experiment of drawing a point uniformly at random from the unit interval $[0, 1]$. Let Y be the first digit after the decimal point of the chosen number. Explain why Y is discrete and find its probability mass function.

Exercise 1.49. An urn has $n - 3$ green balls and 3 red balls. Draw ℓ balls with replacement. Let B denote the event that a red ball is seen at least once. Find $P(B)$ using the following methods.

(a) Use inclusion-exclusion with the events $A_i = \{i\text{th draw is red}\}$.

Hint. Use the general inclusion-exclusion formula from Fact 1.26 and the binomial theorem from Fact D.2.

(b) Decompose the event by considering the events of seeing a red ball exactly k times, with $k = 1, 2, \ldots, \ell$.

(c) Compute the probability by considering the complement B^c.

Challenging problems

Exercise 1.50. We flip a fair coin repeatedly, without stopping. In Examples 1.16 and 1.22 we have seen that with probability one we will eventually see tails. Prove the following generalizations.

(a) Show that with probability one we will eventually get 5 tails in a row.
Hint. It could help to group the sequence of coin flips into groups of five.

(b) Let $\bar{a} = (a_1, a_2, \ldots, a_r)$ be a fixed ordered r-tuple of heads and tails. (That is, each $a_i \in \{\mathtt{H}, \mathtt{T}\}$.) Show that with probability one the sequence $\bar{a}$ will show up eventually in the sequence of coin flips.

Exercise 1.51. A monkey sits down in front of a computer and starts hitting the keys of the keyboard randomly. Somebody left open a text editor with an empty file, so everything the monkey types is preserved in the file. Assume that each keystroke is a uniform random choice among the entire keyboard and does not depend on the previous keystrokes. Assume also that the monkey never stops. Show that with probability one the file will eventually contain your favorite novel in full. (This means that the complete text of the novel appears from some keystroke onwards.)

Assume for simplicity that each symbol (letter and punctuation mark) needed for writing the novel can be obtained by a single keystroke on the computer.

Exercise 1.52. Three married couples (6 guests altogether) attend a dinner party. They sit at a round table randomly in such a way that each outcome is equally likely. What is the probability that somebody sits next to his or her spouse?

Hint. Label the seats, the individuals, and the couples. There are $6! = 720$ seating arrangements altogether. Apply inclusion-exclusion to the events $A_i = \{i\text{th couple sit next to each other}\}$, $i = 1, 2, 3$. Count carefully the numbers of arrangements in the intersections of the A_is.

Exercise 1.53. Recall the definition of a degenerate random variable from Example 1.32. Construct an example of a random variable that is degenerate but not a constant function on Ω. Explicitly: define a sample space Ω, a probability measure P on Ω, and a random variable $X : \Omega \to \mathbb{R}$ such that X satisfies $P(X = b) = 1$ for some $b \in \mathbb{R}$ but $X(\omega)$ is not the same for all $\omega \in \Omega$.

Exercise 1.54. Verify the remark under Definition 1.33 for discrete random variables. Let X be a random variable with countably many values and define

$Q(A) = P\{X \in A\}$ for $A \subset \mathbb{R}$. Show that Q satisfies the axioms of probability. (Note that the sample space for Q is $\mathbb{R}$.)

Exercise 1.55. Show that it is not possible to choose a uniform random number from the whole real line. (In other words, we cannot define a probability on the real numbers that can be considered uniform.)

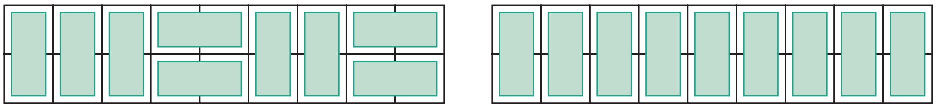
Exercise 1.56.

(a) Put k balls into n boxes at random. What is the probability that no box remains empty? (You may assume that balls and boxes are numbered and each assignment of k balls into n boxes is equally likely.)
(b) Using the result of part (a) calculate the value of

$$\sum_{j=1}^{n} (-1)^{j-1} j^k \binom{n}{j}$$

for $k \le n$.

Exercise 1.57. We tile a 2×9 board with 9 dominos of size 2×1 so that each square is covered by exactly one domino. Dominos can be placed horizontally or vertically. Suppose that we choose a tiling randomly so that each possible tiling configuration is equally likely. The figure below shows two possible tiling configurations.



(a) What is the probability that all the dominos are placed vertically?
(b) Find the probability that there is a vertical domino in the middle of the board (at the 5th position).

Exercise 1.58. Recall the setup of Example 1.27. Find the probability that exactly ℓ people out of n receive the correct hat, and find the limit of this probability as $n \to \infty$.

Hint. Multiply (1.23) by $n!$ to get this counting identity:

$$\#(\text{arrangements where no person among } n \text{ gets the correct hat}) = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}. \tag{1.34}$$

Find the probability that individuals $i_1, i_2, \dots, i_\ell$ and nobody else gets the correct hat. Add over all $i_1 < i_2 < \cdots < i_\ell$. For the limit you should get

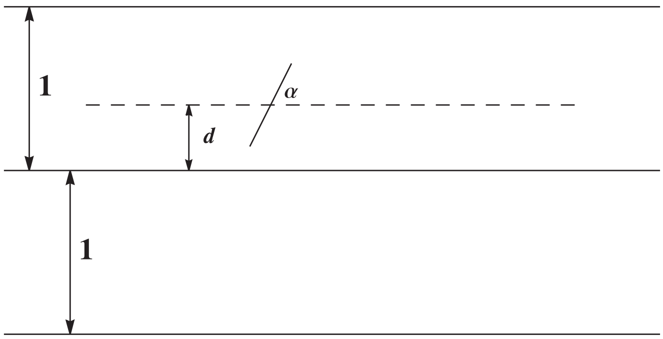
$$\lim_{n\to\infty} P\{\text{exactly } \ell \text{ people out of } n \text{ receive the correct hat}\} = \frac{e^{-1}}{\ell!}.$$

This probability distribution defined by $p(\ell) = e^{-1}/\ell!$ for nonnegative integers ℓ is a particular case of the *Poisson distribution*. We meet it again in Chapter 4.

Exercise 1.59. (Buffon's needle problem) Suppose that we have an infinite grid of parallel lines on the plane, spaced one unit apart. (For example, on the xy-plane take the lines $y = n$ for all integers n.) We also have a needle of length ℓ, with $0 < \ell < 1$. We drop the needle on the grid and it lands in a random position. Show that the probability that the needle does not intersect any of the parallel lines is $2\ell/\pi$.

Hint. It is important to set up a probability space for the problem before attempting the solution. Suppose the grid consists of the horizontal lines $y = n$ for all integers n. The needle is a line segment of length ℓ. To check whether the needle intersects one of the parallel lines we need the following two quantities:

- the distance d from the center of the needle to the nearest parallel line below,
- the angle α of the needle measured from the horizontal direction.



We have $d \in [0, 1)$ and $\alpha \in [0, \pi)$. We may assume (although the problem did not state it explicitly) that none of the possible pairs (d, α) is any likelier than another. This means that we can model the random position of the needle as a uniformly chosen point (d, α) from the rectangle $[0, 1) \times [0, \pi)$. To finish the proof complete the following two steps.

(a) Describe the event $A = \{$the needle does not intersect any of the lines$\}$ in terms of d and α. You need a little bit of trigonometry.
(b) Compute $P(A)$ as a ratio of areas.

This problem was first introduced and solved by Georges Louis Leclerc, Comte de Buffon, in 1777.

Exercise 1.60. (Continuation of Examples 1.7 and 1.40) Consider the experiment of flipping a coin three times with Ω as in Example 1.7. Suppose you learn the outcome of the first flip. What is the collection of events $\mathcal{F}_1$ that represents the information you possess? Suppose you learn the outcome of the first two flips. What is the collection of events $\mathcal{F}_2$ that represents the information you possess?

This exercise suggests that increasing collections of events $\mathcal{F}_k$, $k = 1, 2, 3, \ldots$, can represent the passage of time, as we learn the outcomes of more experiments. This idea is central for modeling random processes that unfold over time.

Exercise 1.61. Use Fact 1.39 and de Morgan's law to show that if $B_1 \supseteq B_2 \supseteq B_3 \supseteq \cdots \supseteq B_n \supseteq \cdots$ is a sequence of nested decreasing events then

$$\lim_{n\to\infty} P(B_n) = P(\cap_{k=1}^{\infty} B_k). \tag{1.35}$$

Exercise 1.62. Using Fact 1.39 to take a limit, deduce the infinite version of Exercise 1.43: for any sequence of events $\{A_k\}_{k=1}^{\infty}$,

$$P\left(\bigcup_{k=1}^{\infty} A_k\right) \le \sum_{k=1}^{\infty} P(A_k). \tag{1.36}$$

Exercise 1.63. Recall the properties of a σ-algebra from Section 1.6. Consider the experiment of rolling a die which has sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$.

(a) What is the smallest collection of events $\mathcal{F}$ satisfying the properties of a σ-algebra?

(b) Suppose that the sets $\{1, 2\}$ and $\{2, 3, 4\}$ are in the collection of events $\mathcal{F}$. Now what is the smallest collection of events $\mathcal{F}$ satisfying the properties of a σ-algebra?